AI SYSTEM DOCUMENTATION
Version 1.0

AI METHODOLOGY

System Architecture · Agents · Protocol · Governance

Framework / Multi-Agent Structure / Operational Logic

KY Song · Seoul · 2026

“Agent 사고 방식 + 엔지니어링 방법론” 을 합친 구조의 방법론을 정의 하였습니다.

1	2	3	4	5	6
Vision & Role 정의	**Agent Workflow 설계**	**Capability 구축**	**Simulation 테스트**	**Deploy & Monitoring**	**Evolution Loop**
AI Agent가 왜 존재하는지, 그리고 각 Agent가 어떤 책임(R&R)을 가지는지 결정하는 단계입니다.	Agent들이 어떤 순서와 규칙으로 협업하는지 설계하는 단계.	Agent가 실제로 일할 수 있게 만드는 단계. (기술구현 단계)	Agent끼리 대화하면서 시스템을 검증하는 단계.	Agent를 실제 환경에서 실행하고 행동을 관찰하는 단계.	Agent가 실제로 실행된 이후의 로그와 결과를 분석해서 Prompt / Workflow / Tool / Memory를 개선하는 반복 운영 프로세스 단계
✓ Problem 정의 ✓ Agent Role 설계 ✓ Human-in-the-loop 정의	✓ Workflow 구조 작성 ✓ Decision Rule 정의 ✓ Tool Flow 설계	✓ Prompt 설계 ✓ Tool 연결 ✓ Memory 설계 ✓ Skill 정의	✓ Agent Conversation Simulation ✓ Failure Scenario 테스트 ✓ Evaluation 기준 설정	✓ Logging ✓ Observability ✓ Guardrail 설정	✓ Log 분석 ✓ Prompt 개선 ✓ Skill 업데이트

2. R&R 및 산출물 > 1. Vision & Role 정의 단계 / 2. Agent Workflow 단계

Phase	Role & Responsibility		WBS Task		산출물
	Human	Agent			
1. Vision & Role 정의 단계					
1.1 Problem Framing	비즈니스 목표 정의 해결할 문제 승인	요구사항 Draft 생성 시장/사례 분석	1.1.1	Problem Statement 작성	1.1.1 Problem Definition Doc
			1.1.2	Stakeholder Goal 정리	
1.2 Persona Definition	핵심 Persona 선택	Persona Behavior 분석 Scenario Draft 생성	1.2.1	User Scenario 작성	1.2.1 Persona Spec 1.2.2 Usage Scenario
			1.2.2	Context Requirement 정의	
1.3 Agent Role Mapping	역할 Scope 승인	Role Responsibility Draft	1.3.1	Role Matrix 작성	1.3.1 Agent R&R Matrix
			1.3.2	Authority Boundary 정의	
2. Agent Workflow					
2.1 Interaction Flow Draft	• Workflow 방향 결정	• Flow Diagram 생성	2.1.1	Conversation Sequence 설계	2.1.1 Workflow Diagram
			2.1.2	Turn 구조 정의	
2.2 Decision Gate 정의	• 승인 단계 정의	• Gate Logic 제안	2.2.1	Approval Rule 작성	2.2.1 Decision Rule Spec
			2.2.2	Escalation 정의	
2.3 Tool Invocation Flow	• Tool 권한 승인	• Tool Flow Draft	2.3.1	Tool Usage Mapping	2.3.1 Tool Flow Spec
			2.3.2	Error Handling 정의	

2. R&R 및 산출물 > 3.Capability 구축 / 4.Simulation Test

Phase	Role & Responsibility		WBS Task		산출물
	Human	Agent			
3. Capability 구축					
3.1 System Prompt 설계	Tone & Policy 결정	Prompt Draft 작성	3.1.1	Instruction Layer 정의	3.1.1 System Prompt Spec
			3.1.2	Guardrail 포함	
3.2 Skill & Tool Integration	API Scope 승인	Skill Code 생성	3.2.1	Tool Interface 작성	3.2.1 Skill Library 3.2.2 Tool Config
			3.2.2	Permission 설정	
3.3 Memory Architecture	개인정보 정책 결정	Memory 구조 제안	3.3.1	Context Storage 정의	3.3.1 Memory Spec
			3.3.2	Retrieval Rule 설정	
4. Simulation Test					
4.1 Agent Role-play Simulation	• Test Scenario 승인	• Simulation 실행	4.1.1	Multi-Agent Test Run	4.1.1 Simulation Log
			4.1.2	Conversation Log 생성	
4.2 Failure Scenario Test	• Failure 기준 정의	• Error Pattern 분석	4.2.1	Loop Failure Test	4.2.1 Issue List
			4.2.2	Role Conflict Test	

2. R&R 및 산출물 > 5. Deploy & Monitoring / 6. Evolution Loop

Phase	Role & Responsibility		WBS Task		산출물
	Human	Agent			
5. Deploy & Monitoring					
5.1 Logging Setup	Log Policy 승인	Logging Config 생성	5.1.1	Message Trace 저장	5.1.1 Logging Config
			5.1.2	Tool Log 기록	
5.2 Observability 구축	KPI 선택	Metric Report 생성	5.2.1	Monitoring Dashboard 구성	5.2.1 Monitoring Dashboard
			5.2.2	Alert Rule 설정	
6. Evolution Loop					
6.1 Trajectory Log 수집	• 분석 대상 선정	• Log Aggregation	6.1.1	Execution Data 수집	6.1.1 Trajectory Dataset
			6.1.2	Failure Tagging	
6.2 Behavior Evaluation	• 개선 우선순위 결정	• Evaluation Report 생성	6.2.1	Role Behavior 분석	6.2.1 Evaluation Report
			6.2.2	Tool Usage 평가	
6.3 Prompt & Workflow Adjustment	• 변경 승인	• Patch Draft 작성	6.3.1	Prompt 수정	6.3.1 Updated Prompt 6.3.2 Patch Note

3. Human-AI RACI Matrix

# RACI Definition
R = Responsible (실행 주체)
A = Accountable (최종 책임/승인)
C = Consulted (협의)
I = Informed (공유)

Human Roles:
PM / Architect / UIUX / Frontend / Backend / Test

Agent Roles:
⭑ Orchestrator Agent
UIUX Agent
Frontend Agent
Backend Agent
Test Agent

Dual-Layer RACI Matrix (Human + Agent Visible)

Legend
H-A = Human Accountable (최종 책임)
H-R = Human Responsible (실행 책임)
A-R = Agent Responsible (실행 Agent)
A-S = Supporting Agent

Human Layer	Agent Layer
•PM → 방향/책임 •Architect → 설계 오너 •Test → 품질 오너	•UIUX Agent → Persona/Conversation 실행 •Fronte/Backend Agent → Tool 실행 •Test Agent → Evaluation 실행 •⭑ Orchestrator Agent → 전체 흐름 제어

3. Human-AI RACI Matrix

Vision & Role 정의

Detailed Step	Human						Agent				
	PM	Architect	UIUX	Frontend	Backend	Test	Orchestra	UIUX Agent	Frontend Agent	Backend Agent	Test Agent
Problem Framing	H-A	H-R	C	I	I	I	-	A-S	-	-	-
Persona Definition	H-A	C	H-R	C	I	C	-	A-R	A-S	-	-
Success Metric	H-A	C	C	C	C	H-R	-	-	-	-	A-S
Agent Role Mapping	H-A	H-R	I	I	I	C	-	A-S	A-S	A-S	A-S

Vision 단계에서는 Agent는 주로 Support 역할입니다.

3. Human-AI RACI Matrix

2. Agent Workflow 단계

Detailed Step	Human						Agent				
	PM	Architect	UIUX	Frontend	Backend	Test	Orchestra	UIUX Agent	Frontend Agent	Backend Agent	Test Agent
Interaction Flow Draft	H-A	H-R	H-R	C	C	C	A-S	A-R	A-S	A-S	A-S
Conversation Turn 설계	C	H-R	H-R	C	I	C	A-S	A-R	A-R	-	-
Decision Gate 정의	H-A	H-R	I	I	I	C	☆ A-R	-	-	-	A-S
Loop & Retry Rule	I	H-A	I	I	C	H-R	☆ A-R	-	-	A-S	A-R
Tool Invocation Flow	I	H-R	I	C	H-R	C	A-S	-	A-S	A-R	A-S
Escalation Rule	H-A	H-R	C	H-R	I	C	☆ A-R	-	A-R	-	A-S

Architect = **설계 오너**
Orchestrator Agent = **실제 흐름 실행**

3. Human-AI RACI Matrix

3. Capability 구축

Detailed Step	Human						Agent				
	PM	Architect	UIUX	Frontend	Backend	Test	Orchestra	UIUX Agent	Frontend Agent	Backend Agent	Test Agent
System Prompt 설계	H-A	H-R	C	I	C	C	I	A-R	-	A-S	A-S
Role Prompt 설계	C	C	H-R	I	I	C	I	A-R	-	-	-
Tool Interface	I	H-A	I	I	H-R	C	A-S	-	-	A-R	A-S
Memory Architecture	I	H-A	C	I	H-R	C	I	-	-	A-R	A-S
UI Interaction Schema	I	C	H-A	H-R	I	C	A-S	A-R	☆ A-R	-	-
Guardrail 적용	H-A	H-R	C	C	C	H-R	A-S	-	-	A-S	A-R

Frontend Agent는 UI Schema에서 핵심 실행자입니다.

3. Human-AI RACI Matrix

4. Simulation 테스트

Detailed Step	Human						Agent				
	PM	Architect	UIUX	Frontend	Backend	Test	Orchestra	UIUX Agent	Frontend Agent	Backend Agent	Test Agent
Test Scenario	H-A	C	C	C	C	H-R	I	-	-	-	A-R
Role-play Simulation	I	C	I	A-S	A-S	H-A	☆ A-R	A-S	A-S	A-S	A-R
Role Conflict Test	I	H-R	I	I	C	H-A	☆ A-R	-	-	A-S	A-R
Tool Failure Test	I	C	I	I	H-R	H-A	☆ A-R	-	-	A-R	A-R
Evaluation Report	H-A	C	I	I	C	H-R	A-S	-	-	A-S	A-R

Simulation은 Test + Orchestrator 중심입니다.

3. Human-AI RACI Matrix

5. Deploy & Monitoring

Detailed Step	Human						Agent				
	PM	Architect	UIUX	Frontend	Backend	Test	Orchestra	UIUX Agent	Frontend Agent	Backend Agent	Test Agent
Deployment Pipeline	I	H-A	I	H-R	H-R	C	A-S	-	A-S	A-R	A-S
Logging Setup	I	H-R	I	I	H-R	C	A-S	-	-	A-R	A-R
Observability Setup	C	H-A	I	C	H-R	H-R	A-S	-	A-S	A-R	A-R
Runtime Control	I	H-A	I	C	C	H-R	☆ A-R	-	A-S	A-S	A-R

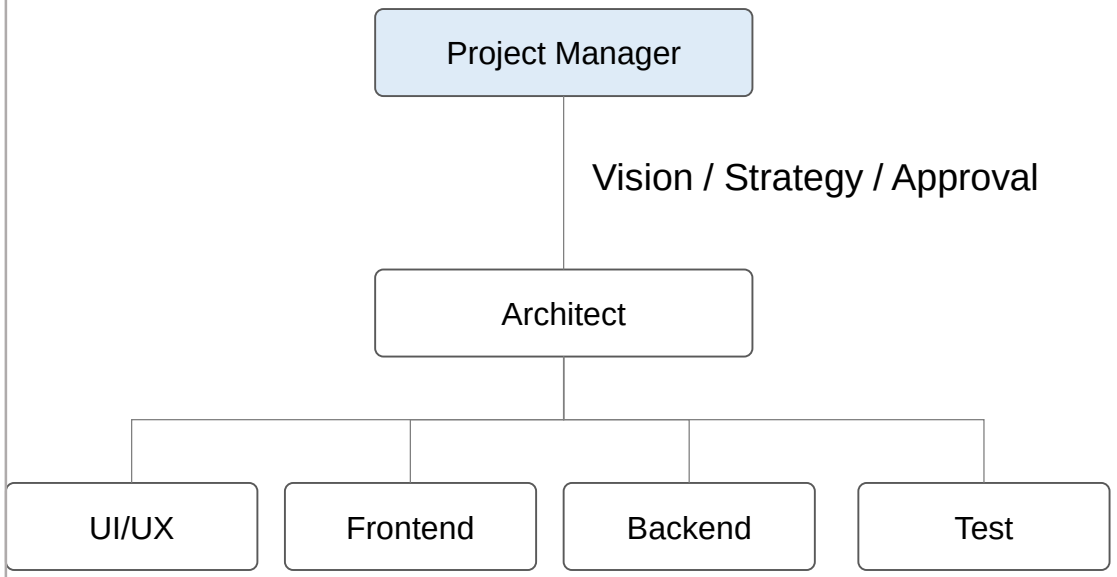
3. Human-AI RACI Matrix

6. Evolution Loop

Detailed Step	Human						Agent				
	PM	Architect	UIUX	Frontend	Backend	Test	Orchestra	UIUX Agent	Frontend Agent	Backend Agent	Test Agent
Trajectory Log 수집	I	C	I	I	H-R	H-R	A-S	-	-	A-R	A-R
Behavior Evaluation	H-A	C	C	I	C	H-R	A-S	-	-	A-S	A-R
Prompt Adjustment	H-A	H-R	C	I	C	C	I	A-R	-	-	A-S
Workflow Patch	I	H-A	I	I	C	H-R	☆ A-R	-	-	A-S	A-R
UI 개선 반영	I	C	H-A	H-R	I	C	A-S	A-R	☆ A-R	-	-

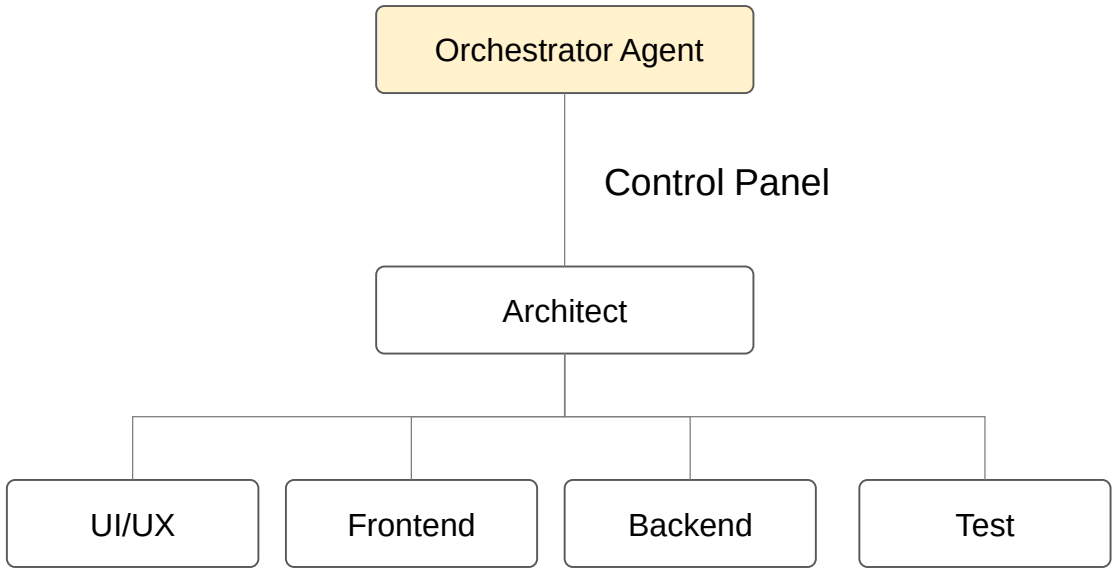
4. AI Agent Dual-Layer Organization Structure (Human + Agent)

HUMAN ORGANIZATION LAYER (Governance / Decision)



UIUX : Persona / Interaction Design
Frontend : UI Engineering / Human Interface Layer
Backend : Tool / API / Runtime
Test : Quality / Evaluation

AGENT ORGANIZATION LAYER (Execution / Automation)



UIUX Agent : Conversation & Persona Execution
Frontend Agent : UI Schema / Layout Generation ☆
Backend Agent : Tool Execution / Integration
Test Agent : Simulation / Evaluation